

SITEX

Spatial Intelligence Platform for Café Site Selection

FEATURES OVERVIEW

A plain-language guide to what SiteX does and how it works

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Section 1: Location Analysis (For Entrepreneurs & Business Owners)

SiteX is built for one goal: helping you decide if a location is a good place to open a café. Instead of guessing or relying on gut feeling, you get a data-backed score for any spot you're considering.

1.1 Choosing a Location on the Map

Drop a Pin, Get a Score

- Open SiteX and you see a live interactive map of your city
- Tap or click any location to drop a pin — a crossroads, a street corner, an empty shopfront
- SiteX immediately analyzes the surroundings and generates a Suitability Score
- Move the pin and the score updates in real time
- No address required — just point and tap

What the Suitability Score Means

- Score ranges from 0 to 100
- Higher scores mean the location has favorable conditions for a café
- You also get a Risk Level: Low, Medium, or High
- The score is not a guarantee of success — it's a data-driven starting point for your decision

How the Score is Calculated

SiteX looks at everything within walking distance of your pin:

- How many competing cafés are already nearby
- Proximity to foot-traffic generators: schools, banks, hospitals, temples
- Road network connectivity — how easy it is for customers to reach you
- Whether nearby competitors create a destination hub or crowd the market
- Your business type (premium, budget, specialty) — so results match your actual situation

1.2 Understanding Your Surroundings

Points of Interest (POI) Layers

- Toggle on/off visual layers showing different POI categories on the map
- Each category appears as a different colored marker:
 - Schools & universities — foot traffic from students
 - Banks & ATMs — indicator of commercial activity
 - Hospitals & clinics — steady visitor flow

- Temples & religious sites — peak-hour congregation patterns
- Existing cafés & restaurants — direct competitors
- See how dense or sparse each category is around your pin
- Use this to understand the neighborhood at a glance before committing

Competitor Proximity View

- Existing cafés within your selected radius are highlighted on the map
- See their names and approximate locations
- SiteX categorizes them: chain café vs. independent, budget vs. premium
- Understand who your direct competition would be before you sign a lease

1.3 Saved Locations & Comparisons

Save Candidate Sites

- Save any pinned location with a custom name (e.g., 'Thamel Corner Plot')
- Return to saved locations at any time — scores are stored
- Add personal notes to each saved location
- Build a shortlist of sites you're seriously considering

Side-by-Side Comparison

- Select two or more saved locations to compare their scores
- See which site scores higher on each factor (foot traffic, competition, road access)
- Helps you make the final call between two shortlisted options

Section 2: Map & Geospatial Features — Understanding Your Surroundings

The map is the heart of SiteX. Everything you see is spatial — laid out on a real map of your city, using actual street and location data.

2.1 Interactive Map Controls

Navigating the Map

- Pinch to zoom in/out on mobile; scroll wheel on desktop
- Drag to pan across the city
- Use the 'My Location' button to center the map on where you are now
- Search by neighborhood name or address to jump to a specific area

Layer Controls

- A sidebar lets you toggle which layers are visible
- Turn on only the layers relevant to your analysis
- Combine layers to see multiple factors at once (e.g., cafés + schools)
- Map stays clean and readable — no information overload

2.2 Heatmaps

Competitor Density Heatmap

- Shows where café concentration is highest across the city
- Red/orange areas = high competition, green/blue = low competition
- Immediately reveals which parts of the city are already saturated
- Helps you spot underserved areas with potential

POI Density Heatmap

- Shows concentration of foot-traffic generators (schools, banks, hospitals)
- Bright areas = high activity zones your café could benefit from
- Overlay with competitor heatmap to find high-traffic, low-competition sweet spots

2.3 Road Network Visualization

Walkability Paths

- See the actual road and path network around your pin
- SiteX traces how customers could walk or drive to your location
- Identifies dead-ends, blocked routes, and barriers (rivers, highways) that cut off access
- A location 100 meters away on the map might be 500 meters by road — SiteX shows this

Isochrone Zones

- An isochrone shows the area reachable within X minutes of walking from your pin
- Choose 5-minute, 10-minute, or 15-minute walking zones
- See how many POIs and competitors fall within each walking zone
- Understand your true catchment area, not just a circular radius

What is an Isochrone?

A regular radius draws a perfect circle around your pin (e.g., 500 meters in all directions).

An isochrone draws the real area you can reach in a set time, following actual roads.

Example: A 10-minute walk east might only reach 400 meters due to a busy highway, but 700 meters west along a pedestrian street. SiteX shows you the real picture.

Section 3: AI Scoring & Explainability — How SiteX Scores a Location

SiteX uses machine learning to calculate suitability scores. But unlike a black-box algorithm, it always explains its reasoning in plain language so you understand exactly why a site got its score.

3.1 The Scoring Engine

How It Works — Step by Step

- You drop a pin on the map
- SiteX measures distances and densities for every relevant POI around that point
- These measurements are fed into a trained machine learning model
- The model outputs a suitability score based on patterns learned from real café success data
- The score and a risk level (Low / Medium / High) appear within seconds

What the Model Has Learned

- The model was trained on data from existing cafés — successful ones and struggling ones
- It has learned which neighborhood characteristics correlate with café performance
- It understands that being near a school is helpful, but being next to five other cafés is risky
- The model gets smarter over time as more data is collected

Graph-Aware Analysis (Advanced)

- SiteX's model understands the road network, not just straight-line distances
- A school 200 meters away via a pedestrian street scores differently than one 200 meters away across a highway
- The model weighs road quality, connectivity, and walkability — not just proximity
- This means locations on main roads with multiple access points score better than identical-distance locations on dead ends

3.2 The Plain-Language Explanation

What Helped Your Score

- After every score, SiteX shows you the top factors that boosted the result
- Example: 'There is a university 180 meters away via a direct pedestrian route — this generates consistent daytime foot traffic'
- Example: 'Only 2 competitor cafés within 500 meters, and neither targets a premium segment'
- Each positive factor is explained in one plain sentence, no jargon

What Hurt Your Score

- SiteX also tells you what dragged the score down
- Example: 'A highway to the west creates a barrier — customers from that direction cannot walk to you'
- Example: 'Three established chain cafés within 300 meters may capture your target customer first'
- Knowing the negatives helps you decide if they can be overcome

Business-Specific Explanations

- If you've told SiteX what type of café you're opening, the explanation adjusts
- A premium café near a budget chain is less threatened than a budget café near the same chain
- Explanations always reflect your specific business model, not a generic average

Why Explanations Matter

A score alone doesn't help you make a decision. Knowing WHY the score is what it is lets you assess whether the negatives are dealbreakers or things you can work around. SiteX is a decision-support tool, not a decision-maker — the explanation is where the real value lies.

Section 4: Business Persona System — Tailoring Results to Your Café

Not all cafés are the same. A high-end specialty roaster and a budget student café have completely different definitions of a good location. SiteX lets you describe your business so every score and recommendation reflects your actual situation.

4.1 Setting Up Your Business Profile

Define What You're Opening

- When you start a new analysis, SiteX asks a few quick questions about your concept:
 - What is your price point? (Budget · Mid-range · Premium)
 - Who is your primary customer? (Students · Professionals · Families · Tourists)
 - What is your café style? (Specialty coffee · Quick-service · Bakery-café · Co-working café)
- These inputs take under a minute to fill in
- You can save multiple profiles if you're evaluating different concepts for the same location

How Your Profile Changes the Score

- The same location can have very different scores for different business types
- Example: A location near a university scores very high for a budget student café but only medium for a premium concept
- SiteX re-weights the scoring factors based on your profile — student foot traffic matters more for some concepts, professional daytime traffic for others

4.2 Persona-Aware Recommendations

Tailored Advice, Not Generic Tips

- Every recommendation SiteX gives is filtered through your business profile
- A premium café gets advice about: nearby income demographics, competitor positioning, ambiance suitability of the area
- A student café gets advice about: proximity to colleges, evening foot traffic, public transport access
- The same tool, completely different outputs — because your situation is different

Pricing & Positioning Guidance

- SiteX suggests a competitive positioning based on who else is in the area
- If three budget cafés are nearby, entering at budget pricing is harder — SiteX flags this
- If there's no premium option in the area but the demographics support it, SiteX highlights the gap

Example: Same Location, Two Different Analyses
Location: Busy corner near a college and a hospital
Budget Student Café Profile:
Score: 82/100 Risk: Low
Reason: High student foot traffic, no direct budget competitors within 300m
Premium Specialty Coffee Profile:
Score: 61/100 Risk: Medium
Reason: Area skews toward budget spending; hospital visitors prefer quick options

Section 5: Competitor Intelligence — Know Your Market

Understanding who's already operating near a site is one of the most important inputs to your decision. SiteX gives you a structured, data-driven view of the competitive landscape.

5.1 Competitor Breakdown

Who's Around You

- SiteX lists all cafés and coffee-serving businesses within your chosen radius
- Each competitor is tagged with:
 - Type: Chain vs. Independent
 - Price tier: Budget · Mid-range · Premium
 - Primary customer: Students · Office workers · Families · General
- This lets you see whether competitors are actually targeting the same customers you are

Enriched Competitor Profiles

- Beyond just 'there's a café here', SiteX gives you context on each one
- Operating hours — are they capturing morning commuters, or evening crowds?
- Approximate capacity and style — tiny grab-and-go vs. large sit-down
- Chain status — chains have marketing budgets and brand loyalty you need to account for

5.2 Synergy vs. Cannibalization Analysis

When Competition is Actually Good

- In some locations, having other cafés nearby is a positive — they create a 'food street' effect
- Customers come to the area specifically because there are multiple options
- SiteX detects this 'destination hub' pattern and flags it as a synergy signal
- Example: A café district where foot traffic is high because people come to browse options

When Competition is a Problem

- In other cases, more cafés simply split the same pool of customers
- This is market cannibalization — everyone earns less because there are too many options
- SiteX detects when the area is at or past saturation point
- You receive a clear warning: 'Market saturation risk — adding another café here likely reduces revenue for all'

The Verdict

- Every location analysis ends with one of three competitor verdicts:
 - Synergy Opportunity: Nearby cafés drive traffic you can capture
 - Healthy Competition: Manageable number of competitors, differentiation is key

- Saturation Risk: Too many similar businesses — entry is high risk

Why This Matters

Most site-selection tools just count how many competitors are nearby.

SiteX goes further — it asks whether those competitors help or hurt you, based on your specific business profile and the actual road network patterns that determine how customers flow through the area.

Section 6: Platform Management (For Administrators & Researchers)

SiteX has a backend admin layer for managing the data pipelines, model performance, and platform health. This section is for the team managing the platform itself.

6.1 Data Pipeline Management

POI Data Ingestion

- Admin dashboard for viewing the current state of all POI datasets
- Run the data refresh pipeline to pull updated POI data for any area
- Review, flag, and correct data quality issues (duplicate entries, incorrect categories)
- Track when each dataset was last updated and its record count

Adding New Cities or Areas

- The automated pipeline can be pointed at a new geographic bounding box
- It fetches, cleans, categorizes, and loads POI data for the new area automatically
- Graph construction (road network + POI nodes) runs automatically after ingestion
- New area becomes available to users once pipeline completes and is validated

Data Quality Monitoring

- Dashboard shows data completeness scores per POI category per area
- Alerts when a category has fewer records than expected (possible data gap)
- Comparison against previous pipeline runs to detect unusual changes

6.2 Model Management

Model Performance Tracking

- View current model version and when it was last trained
- Monitor prediction confidence scores across recent analyses
- Flag analyses where confidence was low (model was uncertain)
- Track model drift — whether predictions are becoming less reliable over time

Retraining Workflow

- Trigger a model retraining run when new data is available
- Compare new model performance against the current deployed model before switching
- Roll back to previous model version if new version underperforms
- All model versions are versioned and stored — nothing is lost

Fallback Model

- If the advanced graph model is unavailable, the system automatically falls back to the XGBoost baseline model
- Users are informed their score used the baseline model, not the full graph analysis
- The baseline model still provides useful scores — just without road-network awareness

6.3 System Health & Audit

System Monitoring

- Real-time view of API response times
- Error rate tracking — how many requests returned errors and why
- Gemini API usage tracker — ensures free-tier limits are not exceeded
- Alert system for any service going down

Audit Logging

- Permanent log of all significant system events
- Tracks: model version changes, data pipeline runs, admin actions
- Searchable and filterable by date, event type, and user
- Cannot be modified or deleted — permanent record for accountability

Section 7: Data & Reporting — Insights & Analytics

7.1 For Users (Entrepreneurs)

Analysis History

- Every location you've analyzed is saved in your history
- Revisit any past analysis — the score, explanation, and map state are all preserved
- See how your saved locations compare to each other over time
- Export any analysis as a PDF report to share with a business partner or investor

Saved Locations Dashboard

- View all your saved candidate sites in one list
- Sort by score, date added, or custom name
- Color-coded by risk level: green (low risk), amber (medium), red (high)
- Archive locations you've ruled out without deleting them

7.2 For Researchers & Urban Analysts

Area-Wide Analysis

- Run a batch analysis across a defined area (e.g., an entire neighborhood)
- Generate a score grid showing suitability across all grid points in the area
- Visualize results as a city-wide heatmap — identify the best and worst zones for café openings
- Useful for urban planners, market researchers, and franchise location teams

Comparative Reporting

- Export side-by-side reports for multiple locations
- Tabular breakdown of each scoring factor per location
- Machine-generated written summary of each location's strengths and weaknesses
- Ready-to-share format for presentations or investor decks

7.3 Scoring Factor Breakdown

Detailed Factor View

- For every score, you can expand a 'Factor Breakdown' panel
- See exactly how much each factor contributed to the final score
- Factors shown:
 - Competitor count and distance
 - Nearest school, bank, hospital distances
 - Road centrality of the location

- Walkable catchment area size
- Synergy vs. cannibalization score
- Persona match score (how well the area fits your business type)
- Bars show positive and negative contribution visually

Section 8: Trust & Accuracy — How We Keep Results Reliable

8.1 Data Sources

Where the Data Comes From

- All POI data is sourced from OpenStreetMap — a free, open, community-maintained dataset
- Road network data comes from the same source — real street maps, not approximations
- The model was trained on real café location data from Kathmandu
- No data is fabricated or estimated — every marker on the map corresponds to a real place

Data Freshness

- POI data is refreshed periodically as the automated pipeline re-ingests new data
- The map always shows when the data for your area was last updated
- New businesses that have not yet appeared in OpenStreetMap may not appear in SiteX
- Users can flag missing or incorrect POIs — these are reviewed and corrected

8.2 Understanding Score Limitations

What SiteX Cannot Know

- SiteX analyzes publicly available geospatial data — it cannot see inside a business
- It does not know actual revenue, footfall counts, or customer satisfaction of nearby cafés
- It cannot predict future changes: a new road, a new competitor, or a neighborhood's gentrification
- Scores reflect current conditions at the time of analysis

Score Confidence Indicator

- Every score comes with a confidence level: High, Medium, or Low
- Low confidence means the model had limited data for that area and the score should be treated with more caution
- High confidence means the model had rich, complete data and the score is more reliable

8.3 Responsible Use

SiteX is a Tool, Not a Verdict

- A high score does not guarantee your café will succeed
- A low score does not mean you should abandon the location — it means you should investigate the flagged factors

- SiteX removes data blindspots — it does not replace your own site visit, market research, and business judgment
- Always visit a shortlisted location in person before making any financial commitment

Our Recommendation

Use SiteX to quickly screen many locations and create a shortlist.

Then do deeper research on your top 2–3 candidates: visit in person, talk to nearby business owners, and verify the factors SiteX flagged.

The best decisions combine data intelligence with human judgment.